

Business Summary Report

Predictive Insights for Collections Strategy | Geldium

1. Summary of Predictive Insights

Dataset context: 500 customers, 19 variables, and 80 delinquent accounts (16.0%).

- Business cardholders** show the highest illustrative segment delinquency rate at 21.3%. Other elevated descriptive segments include Los Angeles customers at 19.6% and unemployed customers at 19.4%. These segment differences are descriptive—not causal—and should be validated before being used for collections decisions.
- Credit Utilization is a leading candidate risk indicator: average utilization was 50.7% among delinquent customers versus 48.9% among non-delinquent customers. Higher utilization may indicate greater credit pressure.
- Debt-to-Income Ratio is another candidate indicator: the average was 30.6% among delinquent customers versus 29.7% among non-delinquent customers. Credit Score is also a key candidate feature, while Income and Loan Balance are relevant measures of repayment capacity and debt exposure. The Task 2 framework recommends validating these relationships through out-of-sample modeling rather than relying on correlation alone.

Top 3 Risk Factors

- 1. Credit Utilization — higher average utilization among delinquent customers suggests greater credit pressure.
- 2. Debt-to-Income Ratio — higher average DTI suggests greater repayment burden.
- 3. Credit Score — important candidate measure of creditworthiness; predictive strength should be validated in modeling.

Key Insight Summary

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Elevated utilization is associated with greater delinquency risk.	Customers in the validated high-risk model band with elevated Credit Utilization.	Credit_Utilization; Debt_to_Income_Ratio; Credit_Score; Income; Loan_Balance.	Prioritize early, human-reviewed outreach and monitor cure/delinquency outcomes.

2. Recommendation Framework

Restated Insight: Customers with elevated Credit Utilization are a higher-priority group for delinquency-risk assessment.

Proposed Recommendation: Run a 6-week targeted collections outreach pilot for customers in the validated high-risk model band who also show elevated Credit Utilization, with human review before intervention.

Specific: Target only the defined high-risk model band and use a supportive early-contact intervention rather than an automatic adverse action.

Measurable: Aim for a 10% relative reduction in 30-day delinquency versus the pilot baseline/comparable control group; also track contact, cure, and escalation rates.

Actionable: Collections can use the model probability and agreed operating threshold to prioritize accounts, then apply approved outreach scripts and document outcomes.

Relevant: The pilot directly supports Geldium's goal of earlier risk identification, better collections prioritization, and reduced delinquency while preserving human oversight.

Time-bound: Operate the pilot for 6 weeks, review results weekly, and complete a formal outcome assessment at the end of the pilot.

Justification and Business Rationale: A focused pilot is more feasible than broad intervention and creates a measurable test of whether predictive prioritization improves collections outcomes. The model should support prioritization—not make automatic adverse decisions—and the operating threshold should reflect business cost and intervention capacity.

3. Ethical and Responsible AI Considerations

Fairness and transparency are essential because collections decisions can materially affect customers.

Fairness Risk 1 — Segment bias: Employment, card type, and location show different descriptive delinquency rates (for example, Business cardholders 21.3% versus Gold 14.4%). These differences may reflect sample composition or structural factors and should not be treated as evidence that a group is inherently riskier.

Mitigation: Validate performance and error rates across relevant groups, monitor fairness and calibration over time, and avoid using segment labels as automatic grounds for adverse action.

Fairness Risk 2 — Data and model quality: The dataset contains only 500 customers and has missing Income (7.8%), Loan Balance (5.8%), and Credit Score (0.4%) values. There are also four Credit Utilization values above 100%. Small samples, missingness, or data anomalies can distort risk estimates.

Mitigation: Use documented training-set-only imputation, investigate anomalous values, validate on representative data, and monitor out-of-sample performance, calibration, stability, and fairness before operational deployment.

Explainability for Stakeholders

- Explain the output as a probability-based risk-prioritization signal, not a statement that a customer will definitely become delinquent.
- Use Logistic Regression as the primary interpretable baseline so coefficients can be reviewed and the contribution of variables can be communicated in plain language.
- Keep a human in the loop: the Task 2 framework explicitly recommends human-reviewed interventions and states that the score should not be used as an automatic adverse action.

Responsible AI Principles

- **Transparency:** document features, thresholds, assumptions, data-quality issues, and intervention rules.
- **Accountability:** assign ownership for model monitoring, threshold changes, exceptions, and customer-impact reviews.
- **Privacy:** use only necessary customer information and avoid exposing raw personally identifiable or unnecessary financial information to GenAI tools.
- **Human oversight:** use predictions to prioritize appropriate support and collections activity, with review before consequential action.